



Indian Institute of Technology Ropar
Department of Mathematics
MA302 - Optimization Methods with Applications
Second Semester of the Academic Year 2025-26

Tutorial sheet - 7

1. Solve the problem

$$\begin{aligned} \text{Max } Z &= 5x_1 + 12x_2 + 4x_3 \\ \text{subject to } x_1 + 2x_2 + x_3 &\leq 5, \\ 2x_1 - x_2 + 3x_3 &= 2, \\ x_1, x_2, x_3 &\geq 0, \end{aligned}$$

(b) Discuss the effect of changing the requirement vector from $\begin{bmatrix} 5 \\ 2 \end{bmatrix}$ to $\begin{bmatrix} 7 \\ 2 \end{bmatrix}$ on the optimum solution.

(c) Discuss the effect of changing the requirement vector from $\begin{bmatrix} 5 \\ 2 \end{bmatrix}$ to $\begin{bmatrix} 3 \\ 9 \end{bmatrix}$ on the optimum solution.

2. A company wants to produce three products A, B and C . The unit profits on these products are Rs.4, Rs.6 and Rs.2 respectively. These products require two types of resources - manpower and material. The following L.P. model is formulated for determining the optimal product mix:

$$\begin{aligned} \text{maximize } Z &= 4x_1 + 6x_2 + 2x_3, \\ \text{subject to } x_1 + x_2 + x_3 &\leq 3, \text{ (manpower)} \\ x_1 + 4x_2 + 7x_3 &\leq 9, \text{ (material)} \\ x_1, x_2, x_3 &\geq 0, \end{aligned}$$

where x_1, x_2, x_3 are the number of products A, B and C produced.

Find the effect of changing the objective function to $Z = 2x_1 + 8x_2 + 4x_3$ on the current optimal product mix.

3. Consider the problem

$$\begin{aligned} \text{Maximize } Z &= 45x_1 + 100x_2 + 30x_3 + 50x_4, \\ \text{subject to } 7x_1 + 10x_2 + 4x_3 + 9x_4 &\leq 1,200, \\ 3x_1 + 40x_2 + x_3 + x_4 &\leq 800, \\ x_1, x_2, x_3, x_4 &\geq 0. \end{aligned}$$

The final simplex tableau:

	c_j	45	100	30	50	0	0	
c_B	Basis	x_1	x_2	x_3	x_4	x_5	x_6	b
30	x_3	$\frac{5}{3}$	0	1	$\frac{7}{3}$	$\frac{4}{15}$	$-\frac{1}{15}$	$\frac{400}{3}$
100	x_2	$\frac{1}{30}$	1	0	$-\frac{1}{30}$	$-\frac{1}{150}$	$\frac{2}{75}$	$\frac{40}{3}$
$Z_j - c_j$		$\frac{25}{3}$	0	0	$\frac{50}{3}$	$\frac{22}{3}$	$\frac{2}{3}$	

If a new variable x_7 is added to this problem with a column $\begin{bmatrix} 10 \\ 10 \end{bmatrix}$ and $c_7 = 120$, find the change in the optimal solution.

4. Consider the following LP problem:

$$\begin{aligned}
 &\text{Maximize} && Z = 3x_1 + 5x_2 \\
 &\text{subject to} && 3x_1 + 2x_2 \leq 18, \\
 & && x_1 + 2x_2 \leq 4, \\
 & && x_2 \leq 6, \\
 & && x_1, x_2 \geq 0.
 \end{aligned}$$

The final simplex tableau:

c_B	Basis	b	x_1	x_2	s_1	s_2	s_3
3	x_1	2	1	0	$\frac{1}{3}$	0	$-\frac{2}{3}$
0	s_2	0	0	0	$-\frac{2}{3}$	1	$\frac{4}{3}$
5	x_2	6	0	1	0	0	1
$Z = 36$			0	0	-1	0	-3

(a) Discuss the effect on the optimal basic feasible solution by adding a new constraint

$$2x_1 + x_2 \leq 8$$

to the given set of constraints.

5. Consider the following linear programming problem:

$$\begin{array}{ll}\text{Maximize} & Z = 3x_1 + 4x_2 + x_3 + 7x_4 \\ \text{subject to} & 8x_1 + 3x_2 + 4x_3 + x_4 \leq 7, \\ & 2x_1 + 6x_2 + x_3 + 5x_4 \leq 3, \\ & x_1 + 4x_2 + 5x_3 + 2x_4 \leq 8, \\ & x_1, x_2, x_3, x_4 \geq 0.\end{array}$$

- (a) Let the variable x_4 be deleted from the LPP. Obtain an optimal solution to the resulting LPP.